BLE options for an industrial data logger

Apollo3 Blue native radio vs. NM180101, u-blox ANNA-B112-02B, ANNA-B412, and Raytac MDBT42TV-AT

Decision	Prototype **ANNA-B112-02B** first for the stated global market set. Prototype **ANNA-B412** in parallel if +105 C operation, longer range, or secure boot is valuable. Its current published approval list omits China. Treat the Apollo3 native radio as a later cost-reduction path. Do not budget NM180101 as a certified shortcut until its exact grants and approved antenna conditions are supplied. [S07] [S09] [S10] [S04] [S05]

Scope and working assumptions

- Board width is fixed at **16 mm**. Board length, layer stack-up, enclosure material, antenna clearance, and required range were not provided; RF fit remains a prototype gate.

- Annual production is **2,000 units**. First release markets used for the estimate: US, Canada, EU, UK, Japan, Australia/New Zealand, China, South Korea, Taiwan, and Brazil.

- Costs are in **USD** except where a distributor price is explicitly quoted in EUR. Engineering is budgeted at **$140 per hour fully burdened**. These are planning estimates, not vendor or test-lab bids.

- One board revision and a defined data-transfer protocol are assumed. A new mobile app, major mechanical changes, hazardous-location approvals, and production tooling are outside the estimate.

At a glance

Path	One-time planning total	Indicative elapsed time	Primary issue
Native Apollo3	$139k-$250k	22-36 weeks	Full RF development and country radio approvals
NM180101	$79k-$142k*	14-24 weeks*	12.8 mm body; external antenna; grants and supply need confirmation
ANNA-B112-02B	$62k-$113k	12-20 weeks	Best published major-market coverage, including China
ANNA-B412	$62k-$113k plus China plan	12-20 weeks for approved markets	China absent from current approval list
MDBT42TV-AT	$59k-$107k*	10-18 weeks*	Fixed serial firmware and security requirements need validation

* Conditional estimate. NM180101 assumes usable grants for the selected antenna and markets. Raytac assumes its supplied UART service satisfies the product requirements. Sources and estimate basis: [S04] [S07] [S10] [S14] [S15] [S17] [S25] [S26]

1. Architecture and physical fit

A module's stated width is not its complete RF footprint. The u-blox internal antenna needs a specified host-PCB antenna strip and reference layout. Raytac requires a ground-free area at the antenna end. The NM180101 exposes a BT_ANT pad and needs an external antenna solution. The installed enclosure and nearby metal affect radiated performance. [S04] [S08] [S11] [S15]

Option	Verified facts	16 mm width assessment	Software fit
Apollo3 native	Existing MCU has BLE controller; Ambiq provides stack and RF design guidance. [S01] [S02] [S03]	No extra module body; reserve an RF matching network and effective antenna location.	Full control over GATT, pairing, OTA and power, with more firmware work.
NM180101	12.8 x 12.8 x 2.4 mm; Apollo3 Blue inside; BT_ANT pin. [S04]	Body leaves only 3.2 mm across board width before antenna routing, clearance or assembly tolerance.	Replacing the existing MCU requires pin and PCB remapping; adding it duplicates processing hardware.
ANNA-B112-02B	6.5 x 6.5 x 1.2 mm; -40 to +85 C; nRF52832; u-connectXpress. [S07]	Plausible with corner/edge reference strip; exact stack-up and RF layout matter. [S08]	Host UART/AT interface, serial service and GATT capabilities.
ANNA-B412	6.5 x 6.5 x 1.2 mm; -40 to +105 C; nRF52833; coded PHY and secure boot. [S09] [S10]	Plausible with u-blox reference strip; exact stack-up and RF layout matter. [S11]	Host UART/AT, richer range and security features.
MDBT42TV-AT	5.8 x 8.3 x 1.3 mm; nRF52805; UART peripheral serial firmware. [S14] [S15]	Plausible if antenna-end no-ground area is preserved. [S15]	Fast for a basic data pipe; validate pairing, access control, custom services and updates.

2. Power and radio behaviour

- NM180101 publishes approximately 3 mA BLE receive and 7-9 mA transmit across stated powers; u-blox B112 lists roughly 5.4 mA receive / 5.3 mA transmit; B412 and Raytac quote values at different power and operating points. **These figures are not directly comparable as battery-life estimates.** [S04] [S07] [S10] [S15]

- Measure complete-logger current across advertising, connected transfer, sleep and wake-up, including UART signalling and the host MCU. Test range in the final enclosure at the required orientation and distance.

- For a metal enclosure, plan an approved antenna location outside the shielded region. The B112 integration manual explicitly warns that its internal antenna cannot be mounted in a metal enclosure. [S08]

3. Development estimate

The hours below are engineering estimates for embedded BLE, RF layout, validation and production support. They are multiplied by the assumed $140/hour rate. They do not include writing a new phone app. Published chip/module information supports the scope, while the hours reflect a planning model rather than supplier quotes. [S01] [S03] [S04] [S07] [S11] [S15]

Work package	Native	NM180101	u-blox	Raytac
RF/PCB/antenna	90-150 h	60-110 h	40-70 h	35-70 h
Bring-up or host protocol	70-120 h	140-240 h	90-150 h	70-110 h
BLE application / security	170-260 h	Included above and QA	Included above and QA	Fixed service; gap may add work
Power, RF and interoperability QA	110-170 h	90-150 h	70-120 h	65-110 h
Manufacturing test / compliance support	60-100 h	50-100 h	20-50 h	30-60 h
Total planning hours	**500-800**	**340-600**	**220-390**	**200-350**
Engineering cost	**$70k-$112k**	**$48k-$84k**	**$31k-$55k**	**$28k-$49k**

Raytac's base case presumes its preloaded transparent service meets the data logger's needs. Its published command set does not establish all desired security and update behaviour; allow another **80-160 h ($11k-$22k)** if an application-layer solution is needed. [S15]

4. One-time project budget

Path	Engineering	Prototypes	Certification / compliance	Total
Native	$70k-$112k	$4k-$8k	$65k-$130k	**$139k-$250k**
NM180101, valid grants	$48k-$84k	$3k-$6k	$28k-$52k	**$79k-$142k**
u-blox B112/B412	$31k-$55k	$3k-$6k	$28k-$52k	**$62k-$113k**
Raytac, supplied firmware sufficient	$28k-$49k	$3k-$6k	$28k-$52k	**$59k-$107k**

5. Recurring unit cost and payback

- Native: provisional incremental RF parts allowance **$1-$3/unit**, pending antenna and matching choice. This is an engineering allowance, not a distributor quote.

- Observed module listings: B112 **$5.25 at a 500-unit reel**; B412 **$6.62 at 500 units**; NM180101 **EUR 10.15 at 100 units and out of stock**; Raytac **EUR 4.57 for one unit** with limited visible stock. Confirm 2,000-unit pricing and annual allocations. [S06] [S12] [S13] [S16]

- A native design saving roughly $4-$6 per unit versus a small module saves about $8k-$12k annually at 2,000 units. That is unlikely to repay the additional development and country-approval cost quickly; exact payback depends on vendor quotes and the final antenna BOM.

6. Certification structure

A certified module reduces repeat intentional-radio testing only when its exact part, antenna and installation comply with its grant and integration manual. The finished logger still needs applicable host evaluation. ISED says a host with certified modules ordinarily does not need separate intentional-radio certification if all intentional emissions come from those modules, but the host must meet applicable requirements, including RF exposure. The FCC directs integrators to its module guidance. [S19] [S20] [S21]

Budget element	Certified module path	Native Apollo3 path
Bluetooth SIG product qualification	$12k	$12k
Finished-product EMC, applicable safety and RF exposure	$8k-$18k	$8k-$18k
Module RF verification, regional documents/labels/admin	$5k-$15k	$5k-$15k product documentation and regional admin
Pre-scan and retest allowance	$3k-$7k	$3k-$7k
Full intentional-radio testing and approvals	Usually covered when grants remain valid	$35k-$75k across planned markets
Rounded total	**$28k-$52k**	**$65k-$130k**

The Bluetooth SIG fee is a published **$12,000 Adopter first qualification** fee as of 1 March 2026. The brand owner must qualify its own product even when it incorporates a supplier's qualified design; reuse can avoid new Bluetooth testing. The other amounts are planning allowances informed by a lab's published cost range and a historical first-hand multi-market project, not regulator fee schedules or bids for this logger. [S17] [S18] [S25] [S26]

Regional implications

- **US / Canada:** follow module antenna and output-power conditions; perform the final-host emissions work and limited transmitter verification described by the module supplier; prepare FCC/ISED host labels and RF exposure assessment. [S08] [S11] [S20] [S21]

- **EU / UK:** the finished manufacturer owns the conformity assessment, technical file and declaration. A module declaration supports this work but does not replace product-level EMC, safety or other applicable requirements. Great Britain currently recognizes CE marking for qualifying radio equipment. [S22] [S23] [S24]

- **Japan, Taiwan, Korea, Brazil, Australia/New Zealand:** verify the exact module grant and host marking / documentation provisions in the integration manual. Local administration varies by market and may be minimal when an approved module is used correctly. [S07] [S08] [S10] [S11] [S15]

- **China:** B112 and Raytac list an approval; B412's current datasheet does not. Budget a separate B412 China route until u-blox supplies updated evidence. A provisional extra $5k-$15k and schedule allowance is prudent, but requires a local quote. [S07] [S10] [S14] [S15]

7. What the $5k-$15k integration allowance buys

This allowance is for a **module-based logger whose approved antenna design is preserved**. It is not a second full radio certification budget and it should not duplicate the separate finished-product EMC, safety and exposure line.

Task	Deliverable or test	Allowance
Integration review	Evidence of exact module/firmware, approved antenna, gain, reference layout/stack-up, RF power setting and final installation. [S08] [S11]	$1k-$3k
Limited transmitter verification	Operate module in complete logger and check radiated fundamental and out-of-band emissions, including relevant harmonics / restricted bands. u-blox requests sufficient FCC Part 15C verification under ANSI C63.10. [S08] [S11]	$2k-$5k
Labels and documents	Contains FCC ID / Contains IC labels, applicable country marks and notices, EU/UK product documentation and declaration. [S08] [S11] [S21] [S23]	$1k-$3k
Local administration	Check local host listing, representative, submission and label purchase requirements; pay only applicable items. [S08] [S11]	$1k-$4k

8. Test allocation without double counting

Check / test	When needed	Budget line
Antenna/layout conformity and conducted settings review	Every chosen module implementation	Integration allowance
Radiated transmitter fundamental and out-of-band verification (FCC Part 15C)	u-blox calls for sufficient verification on assembled host	Integration allowance
Host digital emissions (FCC Part 15B / applicable Canadian ICES)	Completed logger	Finished-product EMC
EU EMC / immunity, safety and applicable RF exposure assessment	Completed logger; exact standards set by lab	Finished-product EMC / safety / exposure
SAR measurement	Only if use position, power and exemption assessment require it	Exposure line or added scope
Simultaneous-transmission / co-location check	Only if another radio transmits concurrently	Added scope; seek lab quote
Full intentional-radio country tests	Native RF or module grant conditions cannot be reused	Native radio line or added scope
Bluetooth product listing / fee	Every branded Bluetooth product design	Separate Bluetooth SIG line

Sources for this allocation: u-blox explicitly separates final-host Part 15B measurements from Part 15C transmitter verification; ISED specifies host RF-exposure and labelling obligations; Bluetooth SIG sets the product qualification obligation. [S08] [S11] [S20] [S21] [S18]

9. Conditions that change the budget or schedule

- **Unapproved antenna, altered reference layout, higher gain or changed power:** ask the module grantee and lab whether a grant modification or new certification is required. The $5k-$15k integration allowance may no longer cover the work. [S08] [S11] [S19]

- **Metal enclosure or very small ground plane:** functional range can deteriorate even if the module body fits. Validate the antenna position in the assembled product before board release. [S08] [S11] [S15]

- **NM180101 grant uncertainty:** Northern Mechatronics markets it as pre-certified, but its public datasheet does not establish the exact grant IDs, approved antenna conditions and country coverage needed for this estimate. Request written evidence. [S04] [S05]

- **Raytac firmware fit:** its AT version is a peripheral-role serial bridge. The published command set does not prove support for the required access-control and firmware-update scheme. Prototype those functions before choosing it for speed. [S14] [S15]

- **EU security timing:** assess any applicable RED cybersecurity requirements for the intended connection model. The European Commission states the current requirements have applied since August 2025, while principal Cyber Resilience Act obligations begin 11 December 2027. This document does not include a bespoke security conformity project. [S22]

- **Other product approvals:** hazardous-location, intrinsic-safety, battery transport, environmental or industry-specific approvals can require additional work after a radio or enclosure change; none was specified for this logger.

10. Recommended qualification sequence

Stage	Decision evidence	Suggested timing
Supplier package	Obtain exact grant IDs, approved antenna/host instructions, latest firmware, production status, 2,000-unit price and lead-time commitment.	Week 0-2
RF fit prototype	Place B112 and B412 reference layouts at proposed board edge; measure range, radiated performance and current in final enclosure.	Week 1-5
Protocol/security prototype	Demonstrate data extraction, bonding/access controls, reconnect, power states and update path with target phone or gateway.	Week 2-7
Pre-compliance	Book host EMC and limited transmitter check; freeze antenna and regional power settings after passing margin.	Week 7-12
Formal release	Complete test reports, country paperwork, Bluetooth qualification, labels and technical file.	Week 12 onward

11. Decision rule

Choose B112 if the required range and +85 C rating suffice and China is in the first release. Choose B412 if its radio/security/temperature benefits matter enough to fund a China approval route or defer China. Choose Raytac only after validating its fixed firmware against security and product needs. Choose native Apollo3 when verified volume, antenna design or lifetime economics justify the larger NRE and longer schedule. [S07] [S09] [S10] [S14] [S15]

Source register

Sources were consulted on **8 October 2026**. Numbered citations in the report link to these documents. Manufacturer and regulator documents establish technical and legal conditions; distributor pages provide observed price/stock snapshots, which can change. Budget hours and non-fixed certification allowances are explicitly this report's planning estimates.

[S01] Ambiq - Apollo3 Blue content portal. BLE design files, SDK and qualification reports. Open source (contentportal.ambiq.com)

[S02] Ambiq - Apollo3 Blue SoC datasheet. Section 6: BLE subsystem and HCI. Open source (ambiq.com)

[S03] Ambiq - Apollo3 and Apollo4 Family BLE User's Guide. Software stack, Cordio examples and application framework. Open source (contentportal.ambiq.com)

[S04] Northern Mechatronics - NM180101 Datasheet, Rev A.3. pp. 1-2 specifications; p. 7 antenna pin; p. 11 grant language; p. 14 package. Open source (northernmechatronics.com)

[S05] Northern Mechatronics - NM180101 product page. Pre-certified marketing claim and product information. Open source (northernmechatronics.com)

[S06] TOP-electronics - NM180101 distributor listing. Observed price, stock status and lead time. Open source (www.top-electronics.com)

[S07] u-blox - ANNA-B112 Data Sheet, UBX-18011707 R13. pp. 5-7 features; p. 30 country approvals and Bluetooth qualification. Open source (www.mouser.com)

[S08] u-blox - ANNA-B112 System Integration Manual, UBX-18009821 R11. Antenna design; pp. 55-59 regional and host verification; Appendix B layout. Open source (content.u-blox.com)

[S09] u-blox - ANNA-B412 product page. Dimensions, temperature, software and feature summary. Open source (www.u-blox.com)

[S10] u-blox - ANNA-B412 Data Sheet, UBX-21028698 R08. p. 9 antenna interfaces; p. 26 approved markets and Bluetooth design. Open source (content.u-blox.com)

[S11] u-blox - ANNA-B4 System Integration Manual, UBX-21000517 R08. Section 7.4.7 host verification; Appendix B antenna reference designs. Open source (content.u-blox.com)

[S12] Mouser - ANNA-B112-02B listing. Observed reel price and stock snapshot. Open source (www.mouser.com)

[S13] Mouser - ANNA-B412-00B listing. Observed reel price, stock and factory lead time. Open source (www.mouser.com)

[S14] Raytac - MDBT42TV-AT product page. Dimensions, UART firmware and regional approval claims. Open source (www.raytac.com)

[S15] Raytac - MDBT42TV-AT / PAT Data Sheet, Version C. pp. 4-12 AT command set; pp. 18-23 footprint and keepout; pp. 37-51 approvals. Open source (www.raytac.com)

[S16] DigiKey - MDBT42TV-AT listing. Observed single-unit price, stock and marketplace status. Open source (www.digikey.es)

[S17] Bluetooth SIG - Schedule of dues and fees. Effective 1 March 2026; Adopter qualification fee. Open source (www.bluetooth.com)

[S18] Bluetooth SIG - Qualify your product. Brand owner's product submission obligation and design reuse. Open source (www.bluetooth.com)

[S19] FCC - KDB 996369: Modules and module integration. Module integration guidance and host manufacturer obligations. Open source (apps.fcc.gov)

[S20] ISED Canada - RSP-100: Certification of Radio Apparatus. Section 8: host and modular product requirements. Open source (ised-isde.canada.ca)

[S21] ISED Canada - RSS-Gen, Issue 6, July 2026. Sections 4.5 and 9.4.4.2: host integration and labelling. Open source (www.ised-isde.canada.ca)

[S22] European Commission - Radio Equipment Directive overview. RED scope, cybersecurity timing and final product assessment. Open source (single-market-economy.ec.europa.eu)

[S23] European Commission - Manufacturers' CE marking responsibilities. Technical documentation and declaration of conformity. Open source (single-market-economy.ec.europa.eu)

[S24] UK Government - Radio Equipment Regulations 2017: Great Britain. CE recognition and conformity assessment guidance. Open source (www.gov.uk)

[S25] Compliance Testing - FCC Certification Cost. Published lab pricing benchmark, not a quotation for this logger. Open source (compliancetesting.com)

[S26] SparkFun - Getting a Module FCC Certified. Historical first-hand multi-market cost example; contextual only. Open source (news.sparkfun.com)